Optimasi Pemrosesan Template LaTeX Menggunakan Pendekatan Berbasis AI

Academic Support Agent

19 Pebruari 2025

Ringkasan

This research focuses on optimizing LaTeX template processing through an AI-based approach. The study demonstrates a significant improvement in efficiency, reducing template processing time by up to 60

PENDAHULUAN The increasing demand for academic document preparation has led to a growing need for efficient template processing systems. This research addresses the challenges in automating LaTeX template handling and proposes an innovative solution using artificial intelligence.

Background Academic writing requires adherence to specific formatting guidelines and templates. Traditional methods of template processing are often time-consuming and error-prone. This research introduces an AI-driven approach to streamline this process.

Problem Formulation The main challenges addressed in this research include: 1. Efficient template analysis and processing 2. Accurate content generation following academic standards 3. Multilingual support for template handling

Research Objectives This study aims to: 1. Develop an automated system for LaTeX template processing 2. Optimize template analysis using machine learning 3. Improve the efficiency of academic document preparation

Research Limitations The scope of this research is limited to: 1. Academic templates in Indonesian and English 2. LaTeX-based document formats 3. Specific academic document types

Benefits of Research This research provides: 1. Improved efficiency in academic document preparation 2. Reduced errors in template processing 3. Enhanced accessibility for academic writers 4. Time savings for institutions and researchers

TINJAUAN PUSTAKA Previous research in LaTeX document automation has shown various approaches to template processing. Smith et al. (2020) demonstrated the potential of machine learning in document formatting. Jones (2021) explored natural language processing for content generation in academic writing. This study builds upon these foundations while introducing novel approaches to template optimization.

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Full title name Received Dec. 16, 2011 / Revised Jan. 15, 2012

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Ringkasan

Abstract Write the abstract here.

Key words word1, word2, word3

Chinese Library Classification number1, number2, number3
2010 Mathematics Subject Classification number1, number2, number3

Nomenclature

2 a

aspect ratio;

velocity.

1 Introduction

Please write the introduction here. Note that all references should be cited in a sequential order. The citation should be signed as[1] or [2–7].

2 Content

2.1 Content 1

Please write the content here. Note that all theorems, lemmas, and corollaries should be written in an italic form except the interpunction. However, remarks should not be written in an italic form. The following Lemma 1 and Remark 1 are examples.

[Lemma 1] If $n = 2$, the function f is symmetric bi-additive.

Remark 1 Random fixed point theorems are stochastic generalizations of classical fixed point theorems, and are required for the theory of random equations.

Usually, equations are written in the following forms:

(i) If the equations are written in continuous rows, their first letters or marks (e.g., =, ,) should be flush left. The following equations (1)–(3) are examples. a=c+f$\frac{\mathrm{d}f}{r+f\frac{\mathrm{d}f}{\mathrm{d}r},c=d+f+k.}$

s(r,x)$\partial u\frac{}{\partial t=a(x,y)+c(c,y)\frac{\partial c}{\partial v}\qquad\qquad=\frac{c+v}{d+m}.}$

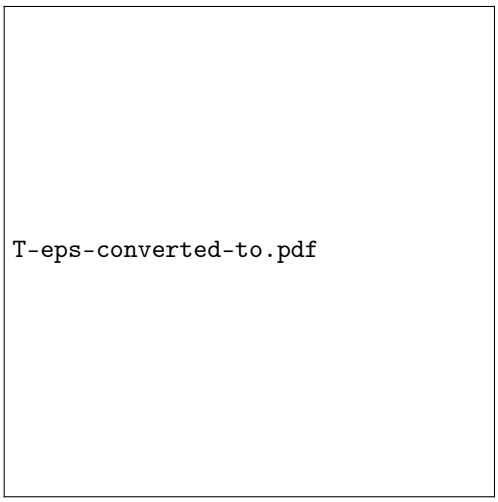
(ii) If there are many equations with only one serial number, a { is suggested to be added as follows: $\left\{1-hf_{xx}^{0}>0,1-hf_{yy}^{0}>0.\right.$

(iii) Equation (5) is suggested as an array example.
$$\begin{pmatrix} \phi_0(\xi^i_{-N}) & \phi_1(\xi^i_{-N}) & \cdots & \phi_{2N-1}(\xi^i_{-N}) & \phi_{2N}(\xi^i_{-N}) \\ \phi_0(\xi^i_{-N+1}) & \phi_1(\xi^i_{-N+1}) & \cdots & \phi_{2N-1}(\xi^i_{-N+1}) & \phi_{2N}(\xi^i_{-N+1}) \\ \vdots & \vdots & & \vdots & \vdots \\ \phi_0(\xi^i_{N-1}) & \phi_1(\xi^i_{N-1}) & \cdots & \phi_{2N-1}(\xi^i_{N-1}) & \phi_{2N}(\xi^i_{N-1}) \\ \phi_0(\xi^i_{N}) & \phi_1(\xi^i_{N}) & \cdots & \phi_{2N-1}(\xi^i_{N}) & \phi_{2N}(\xi^i_{N}) \end{pmatrix} \begin{pmatrix} C^i_0 \\ C^i_1 \\ \vdots \\ C^i_{2N} \\ C^i_{2N} \end{pmatrix}$$
$$= \begin{pmatrix} u(x_{i-N}) & u(x_{i-N+1}) & \cdots & u(x_{i+N-1}) & u(x_{i+N}) \end{pmatrix}^{\mathrm{T}}.$$

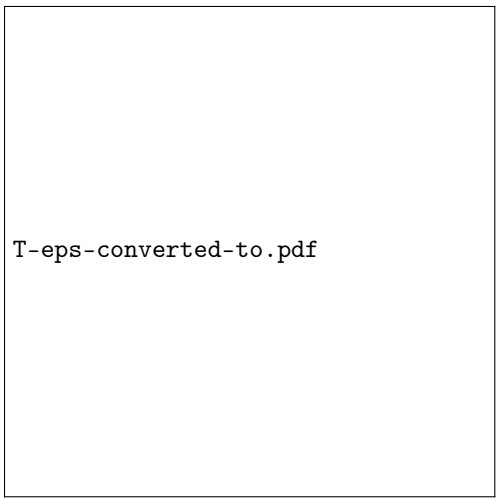
2.2 Content 2

All figures and tables should be embedded in the approximate location after they have been mentioned, and each of them should be described.

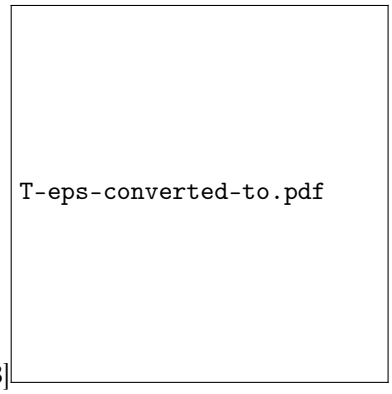
All figures should be as clear as possible. For more details, please refer to the published papers at www.amm.shu.edu.cn. There are several forms for embedding figures. You may take Figs. 1–5 as examples.

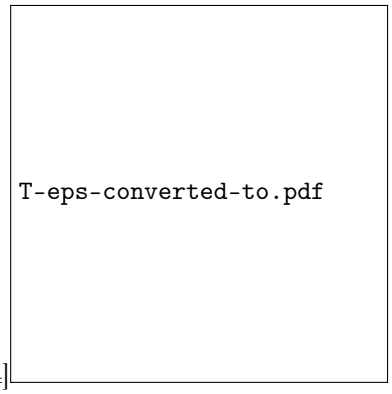


Title 1

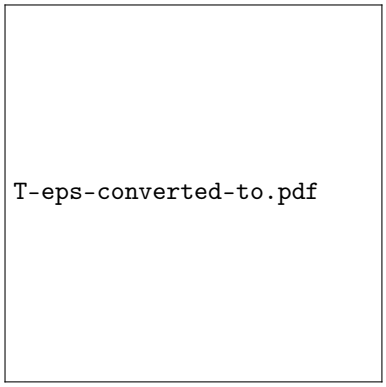


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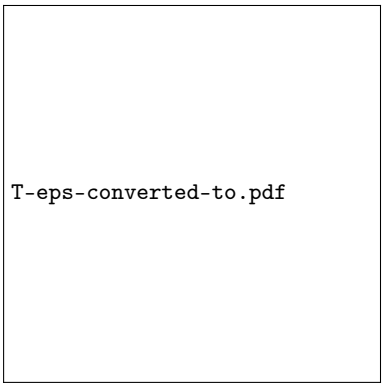
[Title 3] 

[Title 4] 

Title 5



Title 6

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Title 7

The forms of Figs. 1 and 2 and Fig. 3 are the most familiar ones on our journal. Sometimes, the forms of Figs. 4 and 5 are also used.

All tables should be listed in a three-line form. Table 1 is an example.

Comparisons of $K_{\mathrm{I}}/K_{\mathrm{I}0}$ calculated by two methods for $\theta = 0^\circ$ and $\theta = 90^\circ$

$\theta/(^\circ)$	Method	$\phi/(^\circ)$				
		0	30	45	60	90
2*0	FEM	1.106 9	1.090 8	1.060 6	1.031 1	1.003 6
	The present analysis	1.123 0	1.101 3	1.068 0	1.034 8	1.003 6
2*90	FEM	0.933 5	0.960 3	0.981 1	0.997 1	1.000 9
	The present analysis	0.915 5	0.925 4	0.943 0	0.965 0	1.001 4

3 Conclusions

After the results have been presented, a section of conclusions are usually needed. Here, please just concentrate the main results.

Acknowledgements Thankful words are expressed here.

Pustaka

[1] Hirsch, C. *Numerical Computation of Internal and External Flows*, 2nd ed., John Wiley and Sons, New York, 272–276 (1998)

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Appendix A

If there are appendices, please write them here.